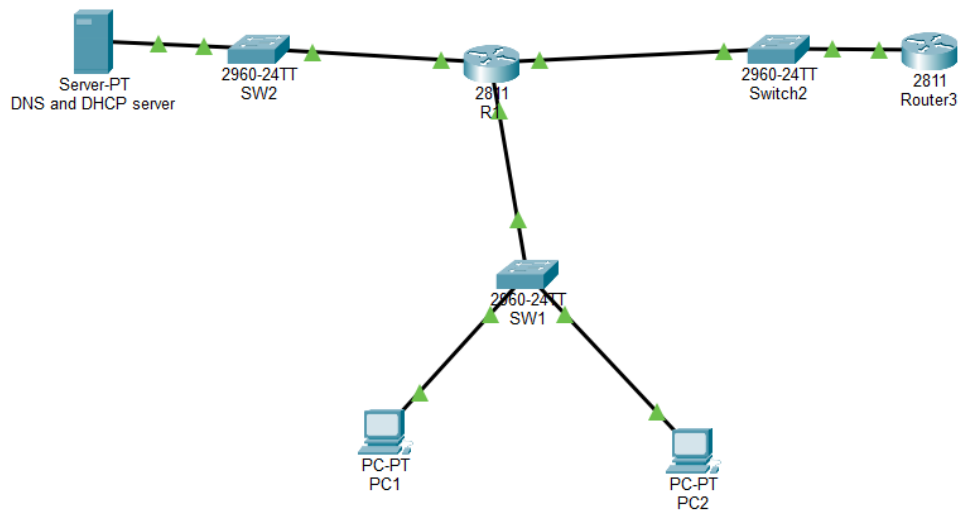


Mpho Edward Makata

Week 10 practical

Topology



97. Setting DHCP Client

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface f0/0
Router(config-if)#ip address dhcp
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#end
```

```
Router#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
FastEthernet0/0    unassigned      YES DHCP    up              up
FastEthernet0/1    unassigned      YES unset    administratively down down
FastEthernet1/0    unassigned      YES unset    administratively down down
FastEthernet1/1    unassigned      YES unset    administratively down down
Vlan1              unassigned      YES unset    administratively down down
Router#
```

98.

```

Router#show ip dhcp pool

Pool INTERNAL-LAN :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 0
  Excluded addresses                : 1
  Pending event                    : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  10.10.10.1         10.10.10.1 - 10.10.10.254    0 / 1 / 254

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp exclude-address 10.10.10.1 10.10.10.20
      ^
% Invalid input detected at '^' marker.

Router(config)#ip dhcp exclude - address 10.10.10.1 10.10.10.20
      ^
% Invalid input detected at '^' marker.

Router(config)#ip dhcp excluded-address 10.10.10.1 10.10.10.20
Router(config)#ip dhcp pool INTERNAL-LAN
Router(dhcp-config)#network 10.10.10.0 255.255.255.0
Router(dhcp-config)#default-router 10.10.10.1
Router(dhcp-config)#dns-server 10.10.20.10
Router(dhcp-config)#exit

```

PC1

IP Configuration	
☒ DHCP	☐ Static DHCP failed. APIPA is being used.
IPv4 Address	169.254.104.167
Subnet Mask	255.255.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

PC2

IP Configuration	
☒ DHCP	☐ Static DHCP failed. APIPA is being used.
IPv4 Address	169.254.212.80
Subnet Mask	255.255.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

```

Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#no ip dhcp Pool Internal_LAN
Router(config)#EXIR
      ^
% Invalid input detected at '^' marker.

Router(config)#EXIT
Router#
%SYS-5-CONFIG_I: Configured from console by console

```

SERVICES	DHCP							
HTTP	Interface		FastEthernet0		Service		On	
DHCP	Pool Name		Internal_LAN					
DHCPv6	Default Gateway		10.10.10.1					
TFTP	DNS Server		10.10.20.10					
DNS	Start IP Address :		10	10	10	2		
SYSLOG	Subnet Mask:		255	255	255	0		
AAA	Maximum Number of Users :		50					
NTP	TFTP Server:		0.0.0.0					
EMAIL	WLC Address:		0.0.0.0					
FTP	Add		Save			Remove		
IoT	Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
VM Management	Internal_LAN	10.10.10.1	10.10.20.10	10.10.10.2	255.255.255.0	50	0.0.0.0	0.0.0.0
Radius EAP	serverPool	0.0.0.0	0.0.0.0	10.10.20.0	255.255.255.0	512	0.0.0.0	0.0.0.0
PRP								

```

Router#
Router#CONF T
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#INTERFACE FASTETHERNET 0/1
Router(config-if)#ip helper-address 10.10.20.10
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

```

100. Because DHCP clients initially send DHCP Discover messages as broadcasts, and routers normally do not forward broadcasts between subnets.